Linux Network Connectivity and Service Inspection

Ubuntu Linux • Oracle VirtualBox • Foundational networking and security

1. Objective

Practice foundational Linux networking and service-inspection techniques in an Ubuntu virtual machine. The lab examines network interfaces, routing, connectivity, listening sockets, and a process associated with a network-discovery service.

2. Lab Environment

- Operating system: Ubuntu Linux (running as a virtual machine)
- Virtualization platform: Oracle VirtualBox
- Network mode observed: VirtualBox NAT
- IPv4 address observed: 10.0.2.15
- Default gateway observed: 10.0.2.2

3. Tools and Commands

Command	Purpose	Observed result
ip addr	Inspect interfaces and assigned IP addresses.	Interface enp0s3 showed 10.0.2.15.
ip route	Inspect routing table and default route.	Default gateway shown as 10.0.2.2.
ping -c 4 10.0.2.2	Test reachability to the gateway.	4 replies; 0% packet loss in captured output.
ping -c 4 1.1.1.1	Test reachability to a public IP address.	4 replies; 0% packet loss in captured output.
ping -c 4 google.com	Test hostname resolution and reachability.	4 replies; hostname resolved and responded in captured output.
sudo ss -tulpen	Inspect TCP/UDP sockets and associated process details.	Several system/application sockets appeared.
ps -fp 3559	Inspect details for process ID 3559.	Output showed a Python process running wsdd.
readlink -f /proc/3559/exe	Identify the executable path for that process.	Output showed /usr/bin/python3.14.

4. Findings and Interpretation

Network configuration and routing

The captured output showed the VM using IPv4 address 10.0.2.15 and a default route through 10.0.2.2. In this VirtualBox NAT configuration, the default gateway is the next hop used to reach destinations outside the VM's local subnet.

Connectivity tests

The three captured ping tests to 10.0.2.2, 1.1.1.1, and google.com each received four replies with 0% packet loss. This demonstrates reachability to those destinations at the time of testing. It does not prove that every network service or route is healthy.

Listening sockets and services

The ss output showed sockets associated with services including local DNS resolution, printing, time synchronization, and network discovery. Port numbers provide context but do not by themselves establish that a service is vulnerable.

- Port 53 is commonly used for DNS. Loopback addresses such as 127.0.0.53 and 127.0.0.54 refer to services local to the VM.
- Port 631 is commonly associated with IPP/CUPS printing.
- Port 5353 is commonly used by multicast DNS (mDNS). A wildcard bind can indicate listening on multiple interfaces; actual reachability depends on protocol, firewall, and network configuration.
- Port 3702 is commonly associated with WS-Discovery. The process inspection showed a Python process running wsdd; this alone is not evidence of malicious activity.

Process investigation

The process inspection command ps -fp 3559 showed a Python process running wsdd. The executable path was reported as /usr/bin/python3.14. A systemctl query for wsdd.service returned that the unit could not be found. This means a service unit with that exact name was not found; it does not by itself prove that the process is suspicious or broken.

5. Security Analysis

This lab demonstrates a basic host-network inventory and connectivity check. Useful analyst habits include checking which interfaces a service binds to, identifying the process associated with a socket, and validating whether the service is expected for the system's purpose. A listening socket is not automatically a vulnerability, and familiar port numbers should be treated as clues rather than proof.

6. Limitations

This was a foundational configuration and connectivity exercise, not a full vulnerability assessment, penetration test, or proof that the VM is secure. The findings reflect the screenshots captured during the lab. No changes were made to firewall rules or services as part of this documented investigation.

7. Conclusion and Next Steps

The exercise provided hands-on practice with Linux network interfaces, routing, ping-based connectivity checks, socket inspection, and process identification. Potential follow-up exercises include reviewing authentication logs, checking service configuration, and capturing traffic with Wireshark.

8. Screenshot Evidence Checklist

- **Figure 1 — Network interface configuration:** Screenshot of ip addr showing enp0s3 and 10.0.2.15.
- **Figure 2 — Routing table:** Screenshot of ip route showing the default route via 10.0.2.2.
- **Figure 3 — Connectivity verification:** Prefer one clear screenshot showing the ping summary for the gateway, 1.1.1.1, and google.com; otherwise use separate cropped images.
- **Figure 4 — Listening sockets:** Screenshot of sudo ss -tulpen with the Local Address:Port and Process columns readable.
- **Figure 5 — Process investigation:** Screenshot of ps -fp 3559 and/or readlink -f /proc/3559/exe.
- **Figure 6 — Firewall status (optional but useful):** Screenshot of sudo ufw status verbose, if it is available and clearly readable.

Suggested Portfolio Description

Performed foundational Linux network troubleshooting in an Ubuntu VM using ip, ping, ss, and ps. Reviewed network addressing and routing, verified connectivity to selected destinations, inspected listening sockets, and investigated a process associated with a network-discovery service. Documented observations and limitations without treating service presence alone as evidence of compromise.

Appendix A — Lab Evidence Screenshots

The following photographs document the commands run in the Ubuntu lab VM. They are supporting evidence for the observations in this report. Because these are photos of the display, some terminal text may be less crisp than a direct screenshot.

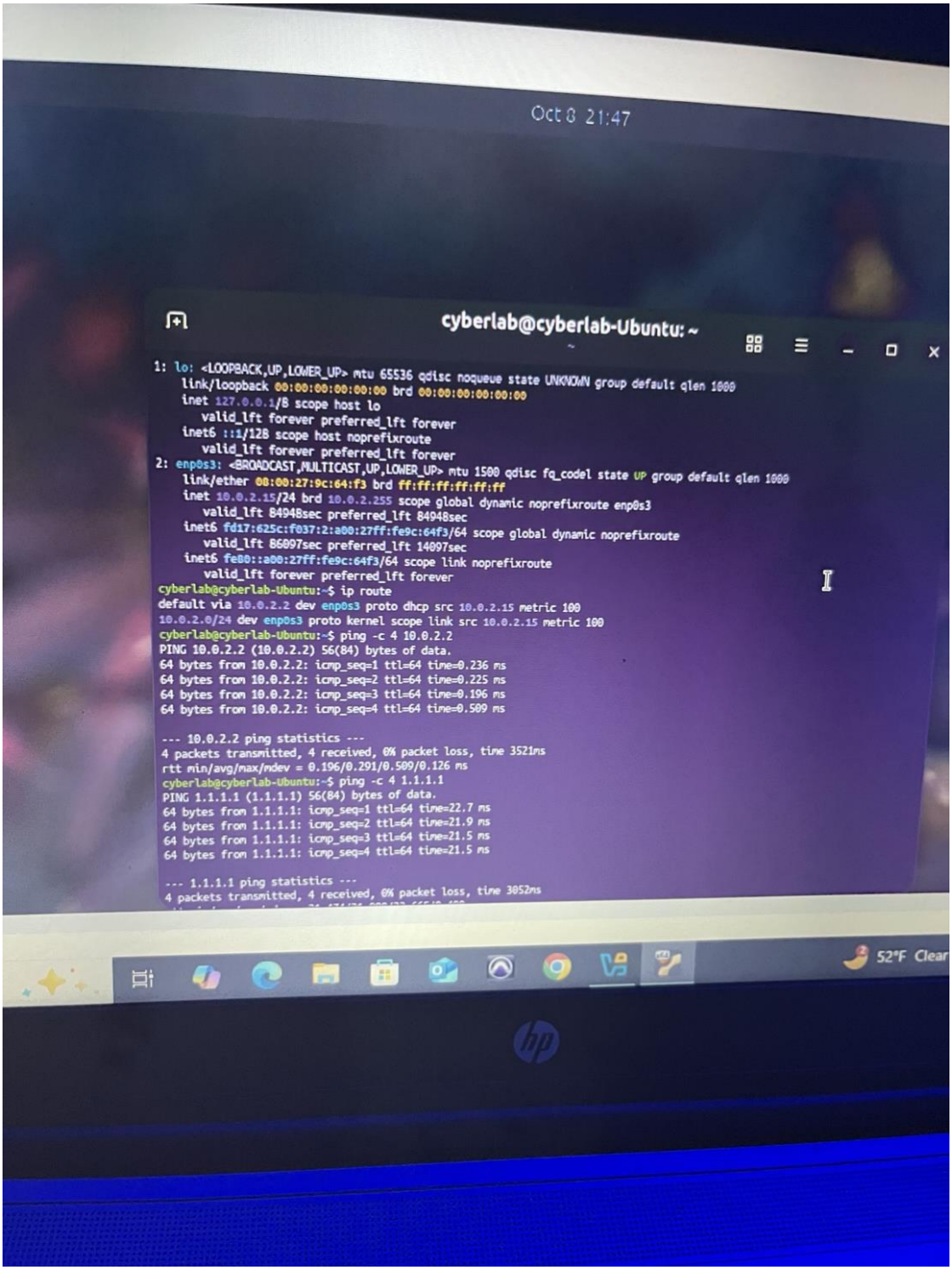


Figure A1. Network interface and route

Shows interface enp0s3 with IPv4 address 10.0.2.15/24 and the default route via 10.0.2.2.

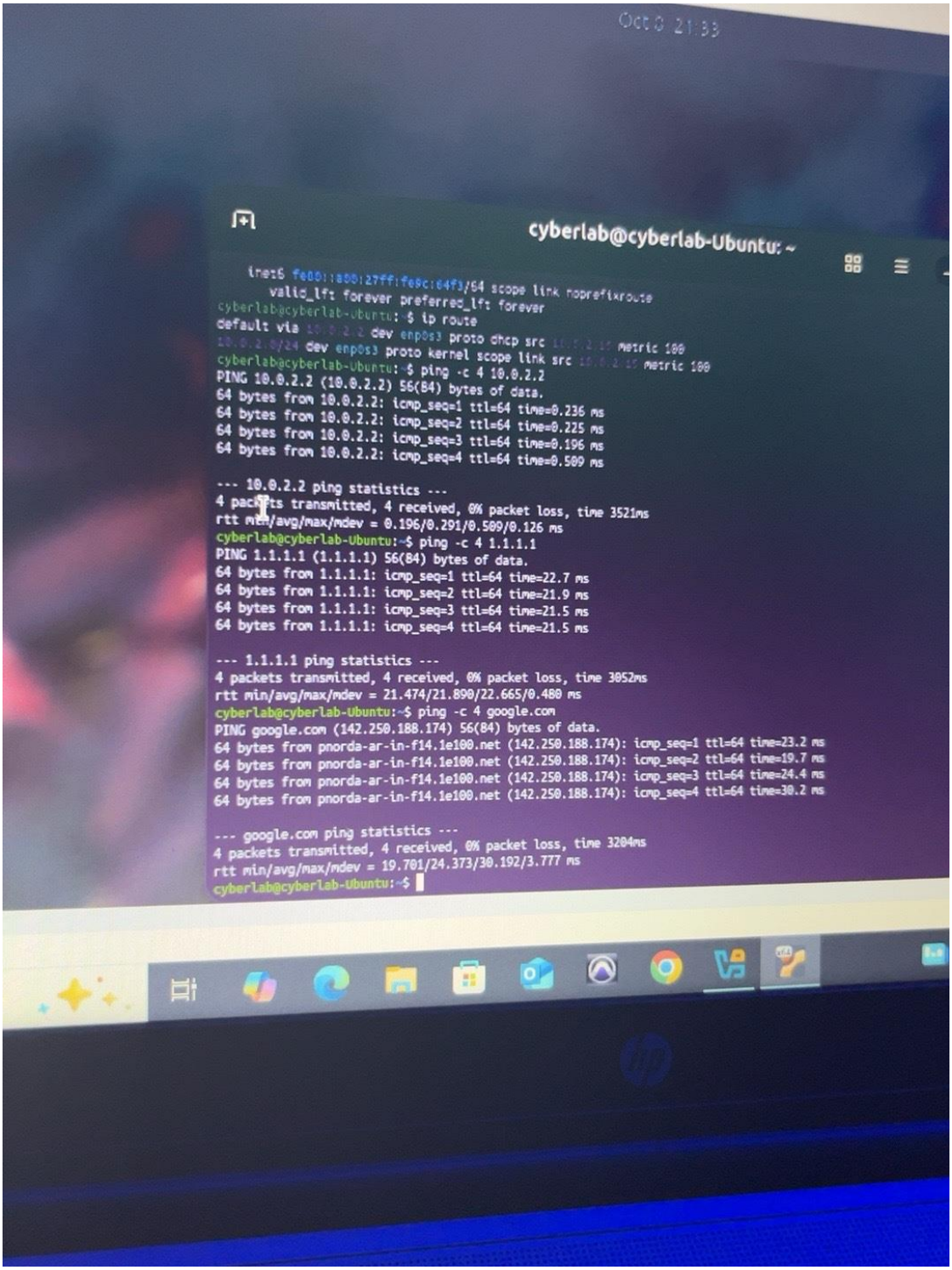


Figure A2. Connectivity tests

Shows successful ping tests to the VirtualBox gateway (10.0.2.2), public IP 1.1.1.1, and google.com; each test reports 4 packets received and 0% packet loss.

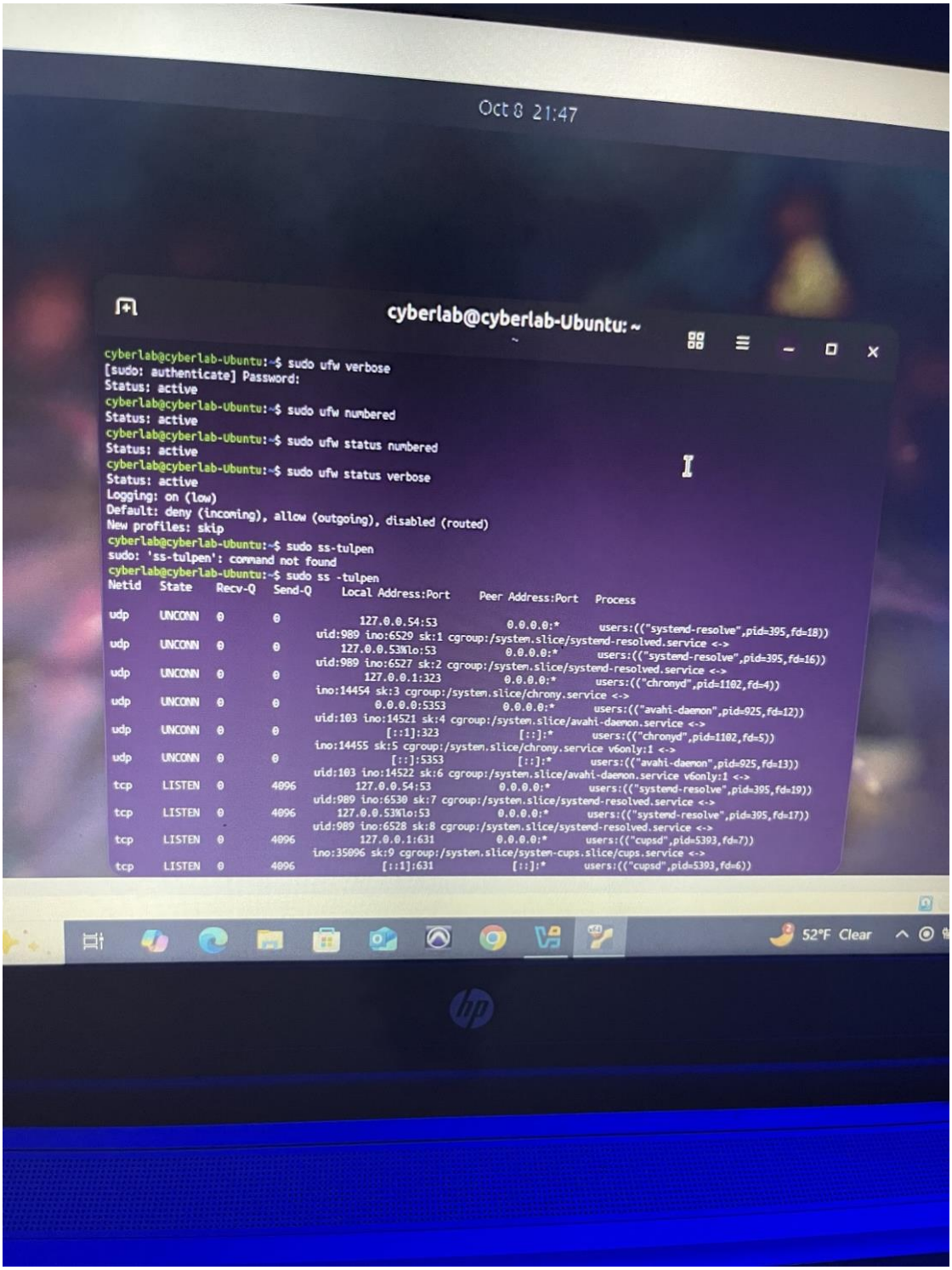


Figure A3. UFW firewall status and socket inspection

Shows UFW active with logging enabled, default incoming deny, and default outgoing allow, followed by the ss socket listing.

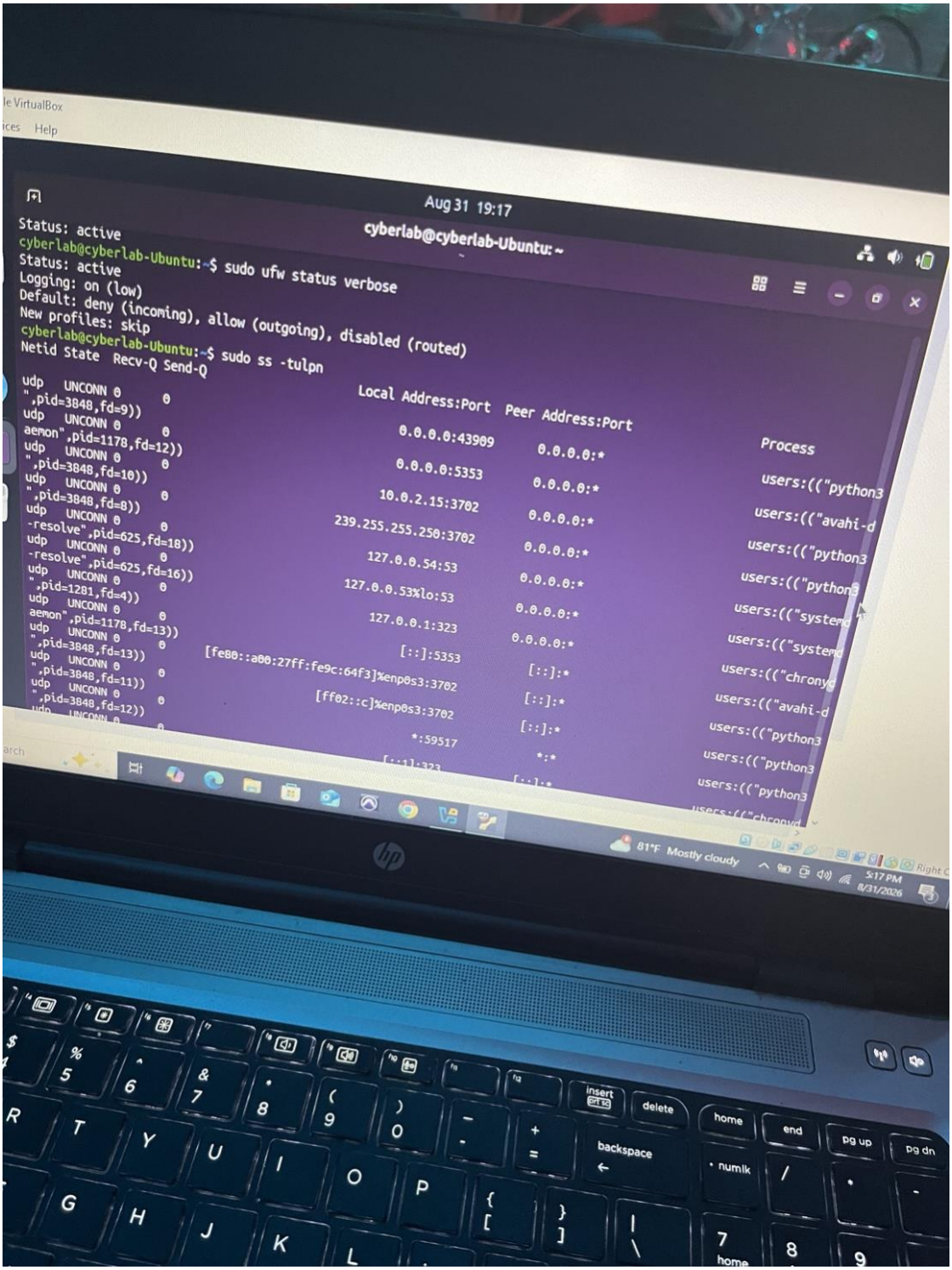


Figure A4. Listening and bound UDP sockets

Shows socket entries including UDP port 5353 and port 3702, including the VM's 10.0.2.15 address and multicast-related addresses.

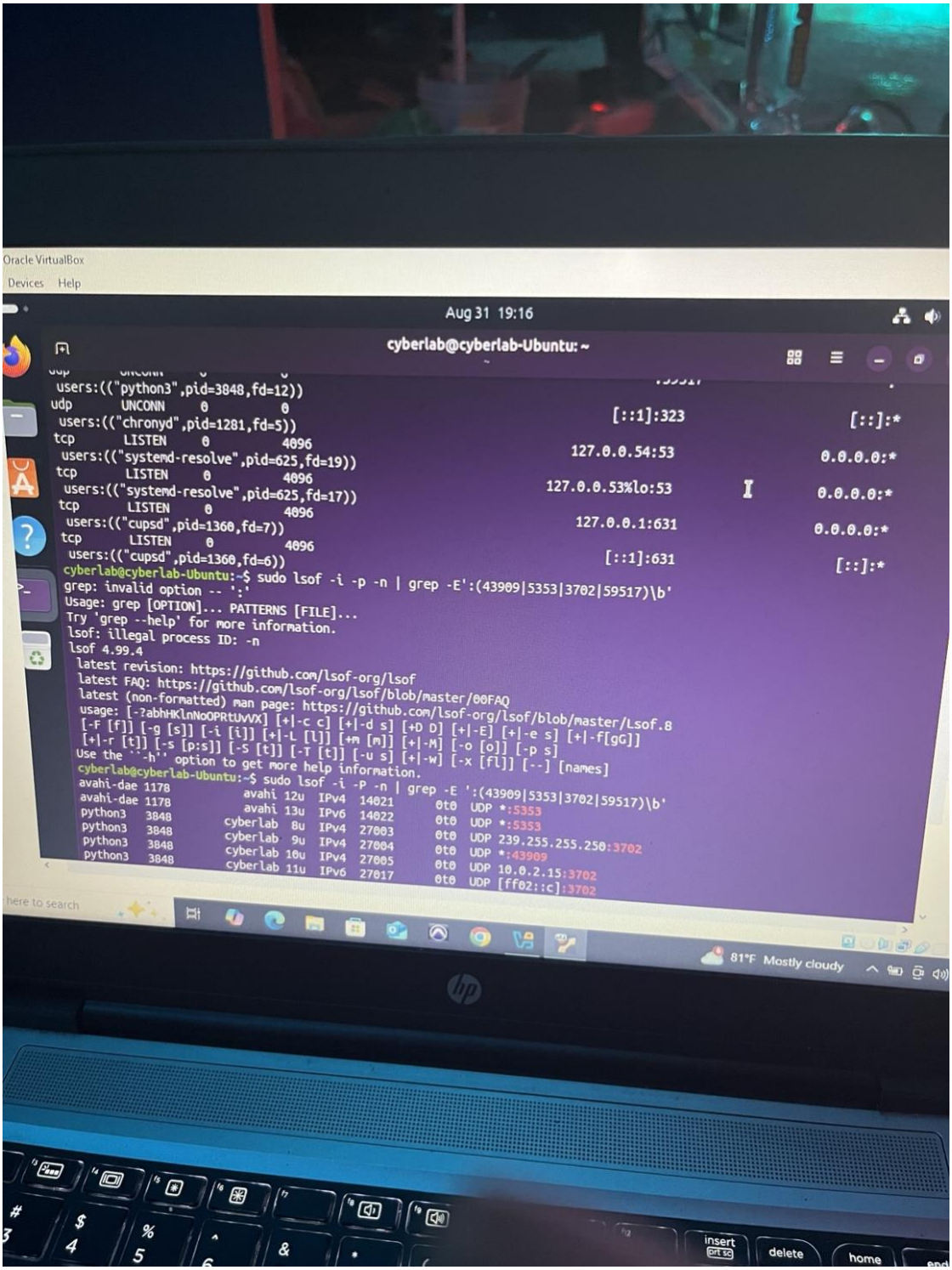


Figure A5. Process-to-port investigation

Shows an lsof query used to associate UDP ports with processes. The earlier malformed command and its error are visible before the corrected command; the corrected output is the useful evidence.

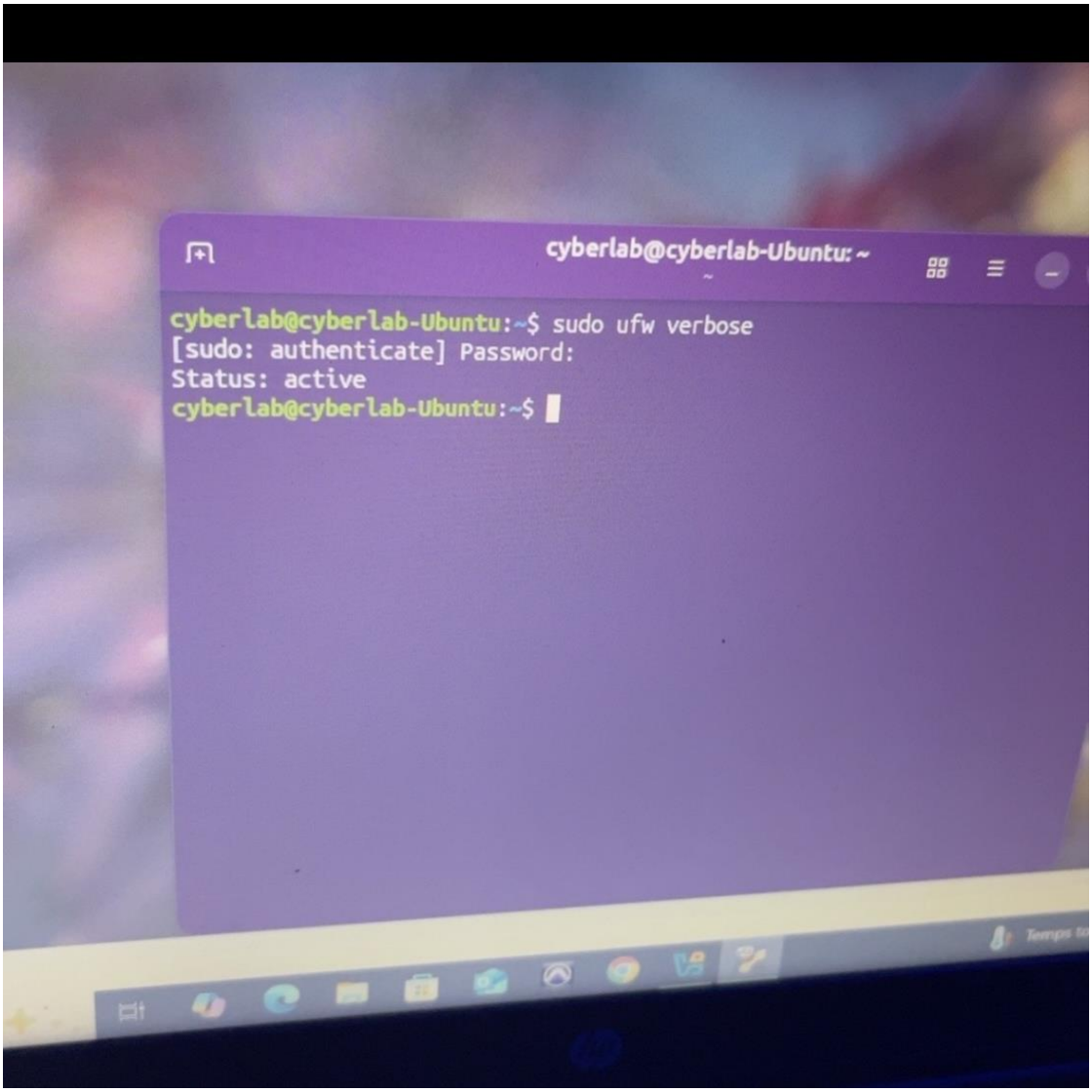


Figure A6. Firewall verification

Shows the UFW status check returning Status: active.

Evidence handling note

For future labs, prefer direct screenshots captured from the VM rather than photos of the monitor. Keep the original image files in the lab's Screenshots folder and use descriptive filenames such as 01-ip-address-and-route.png, 02-ping-tests.png, and 03-ufw-and-sockets.png.